E-Commerce Sales Analysis and Customer Segmentation using SQL and Power BI

Project Objective:

The objective of this project is to analyze e-commerce sales data to identify revenue trends, top-performing product categories, and customer behavior using SQL.

Tools Used:

1. SQL (SQLite using DB Browser for SQLite)
2. Power BI (for visualization)

Dataset:

The dataset used is the Brazilian E-Commerce Public Dataset (Olist), which contains information about orders, customers, products, and payments.

Data Preparation:

A master table was created by joining multiple tables including orders, customers, order items, payments, and products. Only delivered orders were considered for analysis to ensure accuracy.

SQL:

```
-- Create Master Table
CREATE TABLE master_table AS
SELECT
    o.order_id,
    o.customer_id,
    c.customer_unique_id,
    o.order_purchase_timestamp,
    oi.product_id,
```

```
    p.product_category_name,
    oi.price,
    pay.payment_value
FROM orders o
JOIN customers c
    ON o.customer_id = c.customer_id
JOIN order_items oi
    ON o.order_id = oi.order_id
JOIN payments pay
    ON o.order_id = pay.order_id
JOIN products p
    ON oi.product_id = p.product_id
WHERE o.order_status = 'delivered';
```

```
--Data Verificaiton:
    SELECT * FROM master_table LIMIT 10;
```

Output:

	order_id	customer_id	customer_unique_id	order_purch
1	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	7c396fd4830fd04220f754e42b4e5bff	2017-10-0
2	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	7c396fd4830fd04220f754e42b4e5bff	2017-10-0
3	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	7c396fd4830fd04220f754e42b4e5bff	2017-10-0
4	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	af07308b275d755c9edb36a90c618231	2018-07-2
5	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	3a653a41f6f9fc3d2a113cf8398680e8	2018-08-0
6	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	7c142cf63193a1473d2e66489a9ae977	2017-11-1
7	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c	72632f0f9dd73dfee390c9b22eb56dd6	2018-02-1
8	a4591c265e18cb1dcee52889e2d8acc3	503740e9ca751ccdda7ba28e9ab8f608	80bb27c7c16e8f973207a5086ab329e2	2017-07-0
9	6514b8ad8028c9f2cc2374ded245783f	9bdf08b4b3b52b5526ff42d37d47f222	932afa1e708222e5821dac9cd5db4cae	2017-05-1
10	76c6e866289321a7c93b82b54852dc33	f54a9f0e6b351c431402b8461ea51999	39382392765b6dc74812866ee5ee92a7	2017-01-2

The master table combines all relevant data into a single dataset, enabling efficient analysis of sales, customers, and product performance.

Revenue Analysis:

SQL:

```
-- Monthly Revenue Analysis

SELECT

    strftime('%Y-%m', order_purchase_timestamp) AS month,

    SUM(payment_value) AS revenue

FROM master_table

GROUP BY month

ORDER BY month;
```

output:

Month	Revenue
2016-10	61746.94
2016-12	19.62
2017-01	176491.49
2017-02	325782.66
2017-03	505735.83
2017-04	456108.32
2017-05	701313.6
2017-06	585400.98
2017-07	716069.98
2017-08	842689.94
2017-09	996279.59
2017-10	998609.62
2017-11	1548682.69
2018-01	1374064.02
2018-02	1279970.45
2018-03	1435458.33
2018-04	1466607.15
2018-05	1480667.59
2018-06	1285926.11
2018-07	1307228.18

The decline in revenue in the last month is also reflected in the number of orders, indicating a possible drop in demand or incomplete data collection.

Revenue By Categories:

SQL:

```
SELECT
    product_category_name,
    SUM(payment_value) AS revenue
FROM master_table
GROUP BY product_category_name
ORDER BY revenue DESC
LIMIT 10;
```

Output:

cama_mesa_banho	1692714.28
beleza_saude	1620684.04
informatica_acessorios	1549372.59
moveis_decoracao	1394466.93
relogios_presentes	1387362.45
esporte_lazer	1349446.93
utilidades_domesticas	1069787.97
automotivo	833745.67
ferramentas_jardim	810614.93
cool_stuff	744649.32

The analysis shows that categories such as cama_mesa_banho, beleza_saude, and informatica_acessorios generate the highest revenue. These categories represent the core business drivers and indicate strong customer demand in home essentials, personal care, and technology-related products.

RFM Analysis:

Step1>>SQL:

```
create table rfm_base as
SELECT
customer_unique_id,max(order_purchase_timestamp) as last_order_date,
count(distinct order_id) as frequency,
sum(payment_value) as monetory
from master_table
group by customer_unique_id;
select* from rfm_base
limit 10;
```

0000366f3b9a7992bf8c76cfdf3221e2	2018-05-10 10:56:27	1	141.9
0000b849f77a49e4a4ce2b2a4ca5be3f	2018-05-07 11:11:27	1	27.19
0000f46a3911fa3c0805444483337064	2017-03-10 21:05:03	1	86.22
0000f6ccb0745a6a4b88665a16c9f078	2017-10-12 20:29:41	1	43.62
0004aac84e0df4da2b147fca70cf8255	2017-11-14 19:45:42	1	196.89
0004bd2a26a76fe21f786e4fbd80607f	2018-04-05 19:33:16	1	166.98
00050ab1314c0e55a6ca13cf7181fecf	2018-04-20 12:57:23	1	35.38
00053a61a98854899e70ed204dd4bafe	2018-02-28 11:15:41	1	838.36
0005e1862207bf6ccc02e4228effd9a0	2017-03-04 23:32:12	1	150.12
0005ef4cd20d2893f0d9fbd94d3c0d97	2018-03-12 15:22:12	1	129.76

The RFM base analysis shows that most customers have a purchase frequency of 1, indicating a high proportion of one-time buyers. This suggests an opportunity for the business to improve customer retention strategies.

step 2>> SQL:

Recency analysis:

```
CREATE TABLE rfm_recency AS
SELECT
    customer_unique_id,
    julianday((SELECT MAX(last_order_date) FROM rfm_base))
    - julianday(last_order_date) AS recency,
    frequency,
    monetory
FROM rfm_base;
```

	customer_unique_id	recency	frequency	monetory
1	0000366f3b9a7992bf8c76cfdf3221e2	111.169560185168	1	141.9
2	0000b849f77a49e4a4ce2b2a4ca5be3f	114.159143518656	1	27.19
3	0000f46a3911fa3c0805444483337064	536.746921296231	1	86.22
4	0000f6ccb0745a6a4b88665a16c9f078	320.771481481381	1	43.62
5	0004aac84e0df4da2b147fca70cf8255	287.802025462966	1	196.89

Recency was recalculated using the maximum transaction date in the dataset instead of the current date. This provided more meaningful insights, where lower recency values indicate recent customer activity and higher values indicate inactivity.
The analysis shows a significant number of customers with higher recency values, indicating that many customers have not made recent purchases. This highlights an opportunity for re-engagement strategies.

CUSTOMER SEGMENTATION:
SQL:

```
SELECT *,
CASE
```

WHEN recency <= 100 AND frequency >= 2 AND monetory >= 200 THEN 'Champions'

WHEN frequency >= 2 THEN 'Loyal Customers'

WHEN recency > 300 THEN 'At Risk'

ELSE 'Others'

END AS segment

FROM rfm_recency;

Output:

	customer_unique_id	recency	frequency	monetory	segment
1	0000366f3b9a7992bf8c76cfdf3221e2	111.169560185168	1	141.9	Others
2	0000b849f77a49e4a4ce2b2a4ca5be3f	114.159143518656	1	27.19	Others
3	0000f46a3911fa3c0805444483337064	536.746921296231	1	86.22	At Risk
4	0000f6ccb0745a6a4b88665a16c9f078	320.771481481381	1	43.62	At Risk
5	0004aac84e0df4da2b147fca70cf8255	287.802025462966	1	196.89	Others

Customers were segmented based on RFM values into groups such as Champions, Loyal Customers, and At Risk. This helps businesses target different customer groups with appropriate marketing strategies.

9. DASHBOARD ANALYSIS

An interactive dashboard was created using Power BI to visualize key insights from the e-commerce dataset. The dashboard helps in understanding sales performance, product category contribution, and overall business trends.

DashBoard:

Done by power bi

1. Total Revenue

A KPI card is used to display the total revenue generated from all delivered orders. This provides a quick overview of overall business performance.

2. Revenue Trend Over Time

A line chart is used to show how revenue changes over time. The trend indicates a gradual increase in sales across most months, with a slight decline observed in the final period, possibly due to incomplete data.

3. Top Product Categories by Revenue

A bar chart is used to identify the top-performing product categories. Categories such as cama_mesa_banho, beleza_saude, and informatica_acessorios contribute significantly to total revenue.

4. Revenue Distribution by Category

A donut chart is used to show how revenue is distributed across different product categories. This helps in understanding which categories dominate sales.

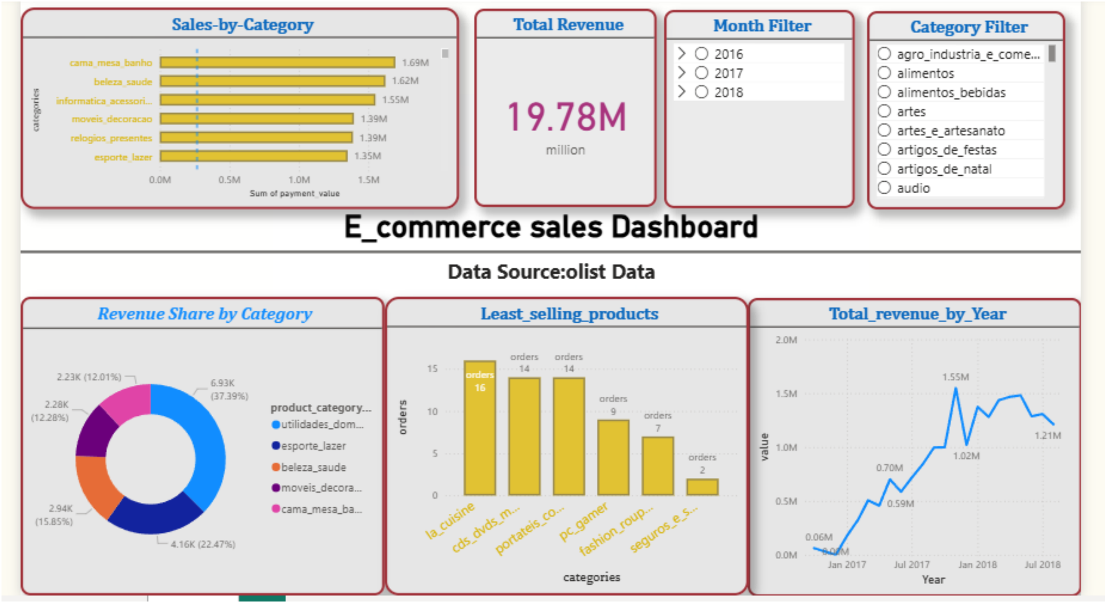
5. Least Performing Categories

A bar chart highlights categories with the lowest number of orders, helping identify underperforming products.

6. Interactive Filters (Slicers)

Slicers are added for month and product category to allow users to filter the data dynamically. This enables better exploration and analysis of specific segments.

Overall, the dashboard provides a clear and interactive way to analyze business performance and supports data-driven decision-making.



Conclusion:

In this project, I analyzed e-commerce data using SQL and performed RFM analysis to understand customer behavior. I identified key revenue-driving product categories and found that most customers are one-time buyers. I also built a Power BI dashboard to present the insights clearly for business decision-making.